

Rock Dynamic Lab Testing

Dynamic rock failure under uniaxial loading, in-situ stress, blast waves, seismic pulses, and excavation disturbance

Multiaxial and path-dependent Hopkinson-bar experiments

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Presentation Overview

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Dynamic Rock-Failure Motivation

Why dynamic rock testing needs more than UCS

Core purpose

Link classical Hopkinson-bar theory, six physics branches, four app loading families, controlled experiments, and damage-model validation in one clear sequence.

Experimental need

Dynamic rock failure is rarely uniaxial. In-situ stress, blast waves, seismic pulses, and excavation disturbance are multiaxial and path dependent.

Analysis need

Every calculation must be traceable from setup, through wave reduction, to stress path, damage, energy density, and validation descriptors.

One platform, four UI families, six physics branches

- X-axis gas-gun train for classical SHPB and coupled triaxial loading.
- Independent Y and Z bar pairs for true triaxial static pre-stress.
- Electromagnetic pulse generators for programmable amplitude, duration, symmetry, and inter-axis delay.
- Current digital-twin pre-stress range: 0–50 MPa; full symmetric XYZ uses one hydrostatic p_0 control.

Physical interpretation

The app shows four top-level families. Modes 4–6 are grouped under *Electromagnetic programmable loading* and selected by EM topology: single-axis, multi-axis one-sided, or symmetric pairs.

Classical Hopkinson-Bar Theory

The equations reduce everything to four checks

1. Wave propagation

Elastic bar-wave mechanics set the usable signal window.

3. Stress path

$\sigma_1, \sigma_2, \sigma_3$ are converted to p , q , and Lode angle θ .

2. Specimen equilibrium

The three-wave method requires transit-time and force-balance checks.

4. Damage and energy

Failure index, stiffness loss, and energy density form the validation layer.

Rate calibration used across the experiments

Updated DIF convention

The analysis presets use a literature-reference strain-rate scale of

$$\dot{\epsilon}_{\text{ref}} = 10^{-5} \text{ s}^{-1}.$$

Consequence

Worked examples, reduced data, and model comparisons follow the same calibration policy rather than mixing independent strength conventions.

$$\text{DIF} = 1 + C \log_{10} \left(\frac{\dot{\epsilon}}{10^{-5}} \right), \quad \text{DIF} \leq \text{DIF}_{\text{max}}.$$

$\dot{\epsilon}$ is the measured strain rate; C is the fitted rate-sensitivity coefficient; DIF_{max} caps the strength amplification outside the calibrated range.

How the classical Hopkinson bar is modified

Mode family	Classical assumption	Laboratory modification	New physics exposed
Mode 1	One-dimensional X-bar impact	No modification; reference SHPB reduction	Baseline dynamic strength
Mode 2	One driven axis with passive confinement	Traditional confinement chamber, $\sigma_2 = \sigma_3 = p_c$	Pressure strengthening at fixed Lode state
Mode 3	One driven axis plus independent boundaries	Active true-triaxial Y/Z confinement	In-situ stress and Lode-angle effects
Modes 4–6	Gas-gun pulse tied to striker speed and length	Electromagnetic half-sine waves with controlled amplitude, duration, timing, and symmetry	Blast/seismic pulse shape, path dependence, and compaction

Narrative order

Start from the classical SHPB equations, then add confinement, true triaxiality, and finally programmable electromagnetic wave loading.

Mode 1: gas-gun uniaxial SHPB

Use when:

- Baseline dynamic uniaxial strength is the target.
- Published classical SHPB comparison is needed.
- Confinement and Lode-angle effects are outside scope.

Main limitation

Single loading axis, no active confinement.

$$\sigma_x(t) \approx \sigma_{\text{dyn}}(t), \quad \sigma_y = \sigma_z = 0.$$

σ_x is axial specimen stress; σ_{dyn} is the dynamic stress inferred from bar waves.

Mode 1 setup conditions

Material and geometry

Sandstone preset: $E = 15$ GPa, UCS = 80 MPa, $\nu = 0.20$, $\rho = 2650$ kg m⁻³.

Specimen: $50 \times 50 \times 50$ mm.

Loading setup

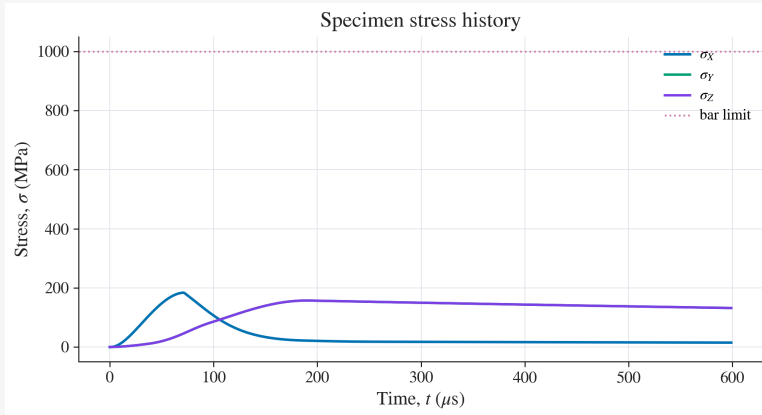
Gas-gun impact along X; striker velocity $V = 20$ m s⁻¹.

Static pre-stress: $\sigma_x = \sigma_y = \sigma_z = 0$ MPa.

Interpretation

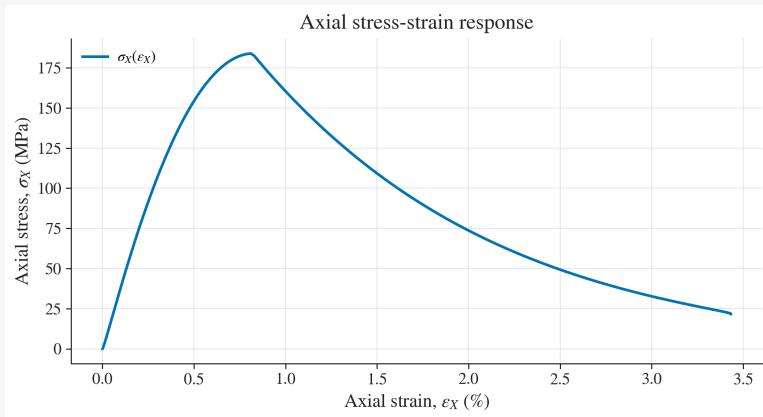
This is the reference unconfined SHPB case used to separate pure axial rate response from confinement and Lode-angle effects.

Mode 1: stress history



- Only the X-axis stress carries the dynamic pulse; lateral stresses remain zero.
- Use this plot to check the baseline pulse duration and peak axial stress before adding confinement.

Mode 1: axial stress–strain response



- The loading branch gives the dynamic uniaxial strength estimate for the sandstone preset.
- No confinement correction should be applied to this baseline curve.

Modified Gas-Gun Hopkinson-Bar Modes

Mode 2: confinement-chamber SHPB

Use when:

- Axisymmetric triaxial confinement is sufficient.
- The specimen is cylindrical or jacketed.
- The stress path can remain at $\theta = 0$.

Main limitation

Lateral pressure is passive; independent σ_2 and σ_3 cannot be imposed.

$$\sigma_2 = \sigma_3 = p_c, \quad p = \frac{\sigma_1 + 2p_c}{3}.$$

p_c is chamber pressure; p is mean pressure;
Mode 2 enforces axisymmetric confinement.

Mode 2 setup conditions

Material and geometry

Sandstone preset: $E = 15$ GPa, UCS = 80 MPa, $\nu = 0.20$, $\rho = 2650$ kg m⁻³.

Specimen: $50 \times 50 \times 50$ mm.

Loading setup

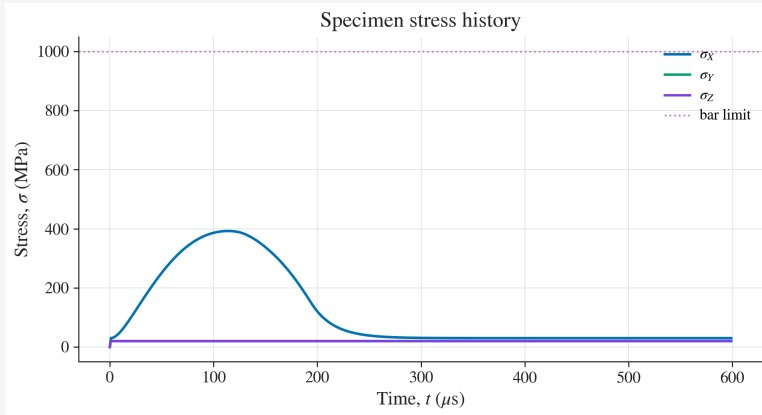
Gas-gun impact along X; striker velocity $V = 20$ m s⁻¹.

Static condition: $\sigma_x = 30$ MPa, chamber pressure $p_c = \sigma_y = \sigma_z = 20$ MPa.

Interpretation

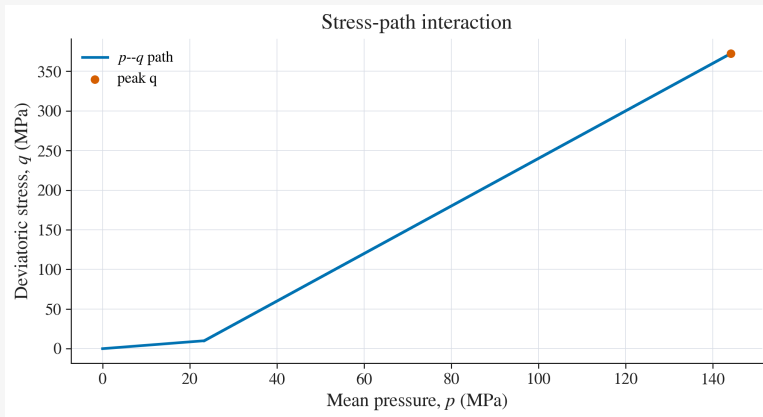
The chamber raises mean pressure but keeps the two lateral principal stresses equal, so the stress path is axisymmetric.

Mode 2: stress history



- The X dynamic pulse rides on top of the static axial pre-stress.
- The lateral stresses remain equal, representing chamber confinement rather than true triaxial control.

Mode 2: stress path



- The p - q path mainly measures confinement strengthening under $\sigma_y = \sigma_z$.
- It is useful for chamber tests but cannot isolate separate Y and Z boundary effects.

Mode 3: Monash gas-gun true triaxial loading

Use when:

- In-situ pre-stress matters.
- σ_2 and σ_3 must be independently controlled.
- Dynamic X loading is combined with true triaxial confinement.

Signature capability

Gas-gun dynamic loading plus active orthogonal bar/cylinder confinement.

$$\sigma_x = \sigma_{x0} + \sigma_{\text{dyn}}(t), \quad \sigma_y = \sigma_{y0}, \quad \sigma_z = \sigma_{z0}.$$

σ_{i0} denotes static pre-stress on axis i before the gas-gun pulse arrives.

Mode 3 setup conditions

Material and geometry

Sandstone preset: $E = 15$ GPa, $UCS = 80$ MPa, $\nu = 0.20$, $\rho = 2650$ kg m⁻³.

Specimen: $50 \times 50 \times 50$ mm.

Loading setup

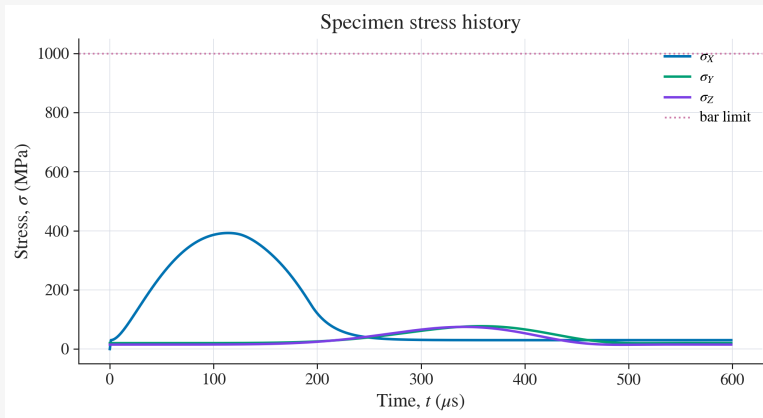
Gas-gun impact along X; striker velocity $V = 20$ m s⁻¹.

Static pre-stress: $\sigma_x = 30$, $\sigma_y = 20$, $\sigma_z = 15$ MPa.

Interpretation

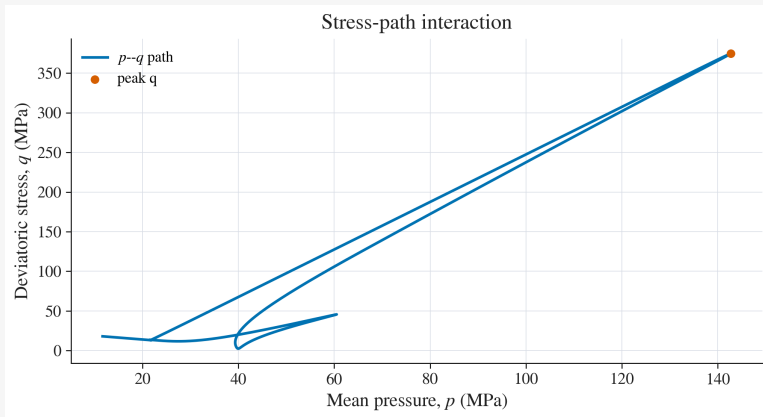
Independent lateral confinement allows the gas-gun result to probe non-axisymmetric stress paths and Lode-angle sensitivity.

Mode 3: stress history



- The dynamic X pulse is combined with unequal static Y and Z stresses.
- This separates true triaxial confinement from the chamber-style $\sigma_Y = \sigma_Z$ assumption.

Mode 3: stress path



- The path moves through a different Lode state because σ_y and σ_z are unequal.
- Compare with Mode 2 to see what changes when lateral stresses are independently imposed.

Electromagnetic Loading Modes

Mode 4: electromagnetic uniaxial impact

Use when:

- A clean half-sine pulse is preferred.
- Pulse amplitude and duration must be tuned independently.
- Rate effects are the main research question.

Design advantage

The EM system decouples peak stress from pulse duration.

$$\sigma_{\text{dyn}}(t) = \sigma_{\text{peak}} \sin\left(\frac{\pi t}{\tau}\right), \quad 0 < t < \tau.$$

σ_{peak} is EM pulse amplitude; τ is pulse duration.

Mode 4 setup conditions

Material and geometry

Sandstone preset: $E = 15$ GPa, $UCS = 80$ MPa, $\nu = 0.20$, $\rho = 2650$ kg m⁻³.

Specimen: $50 \times 50 \times 50$ mm.

Loading setup

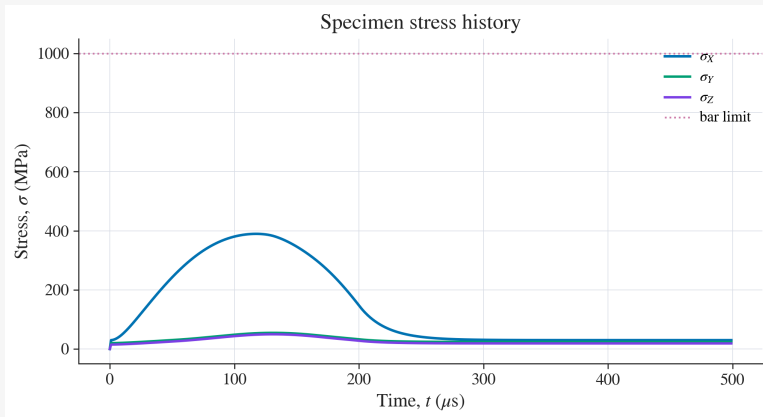
EM half-sine pulse along X; peak 400 MPa, duration 200 μ s.

Static pre-stress: $\sigma_x = 30$, $\sigma_y = 20$, $\sigma_z = 15$ MPa; striker velocity not used.

Interpretation

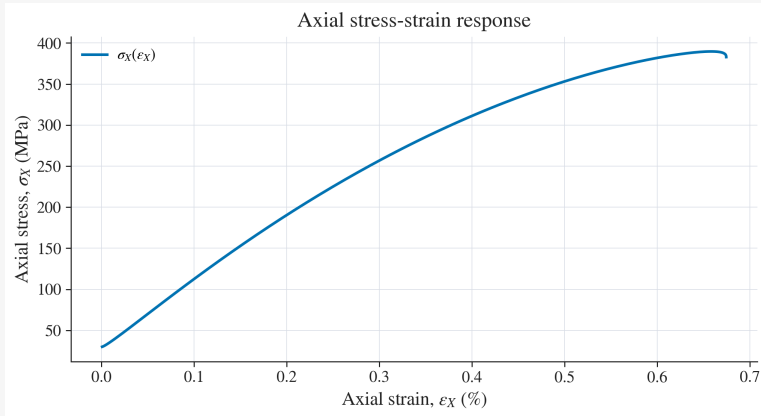
The EM actuator makes pulse amplitude and duration explicit inputs, so rate effects can be changed without changing a gas-gun impact speed.

Mode 4: stress history



- The X-axis stress follows the prescribed half-sine pulse shape.
- Static Y and Z pre-stresses define the initial confinement but do not receive dynamic pulses.

Mode 4: axial stress–strain response



- The branch is controlled by the EM pulse amplitude and duration rather than striker velocity.
- Compare with Mode 1 to separate pulse-shape effects from unconfined gas-gun loading.

Mode 5: electromagnetic asynchronous triaxial impact

Use when:

- Loading order and inter-axis delay are the physics.
- Crack initiation may depend on which axis arrives first.
- Step 2 delay regime and Step 3 stress path are central outputs.

Interpretation warning

A uniaxial constitutive fit is not meaningful for asynchronous triaxial paths.

$$\sigma_i(t) = \sigma_{i0} + A_i g(t - \Delta t_i).$$

A_i is axis- i pulse amplitude; g is the pulse shape; Δt_i is the firing delay.

Mode 5 setup conditions

Material and geometry

Sandstone preset: $E = 15$ GPa, $UCS = 80$ MPa, $\nu = 0.20$, $\rho = 2650$ kg m⁻³.

Specimen: $50 \times 50 \times 50$ mm.

Loading setup

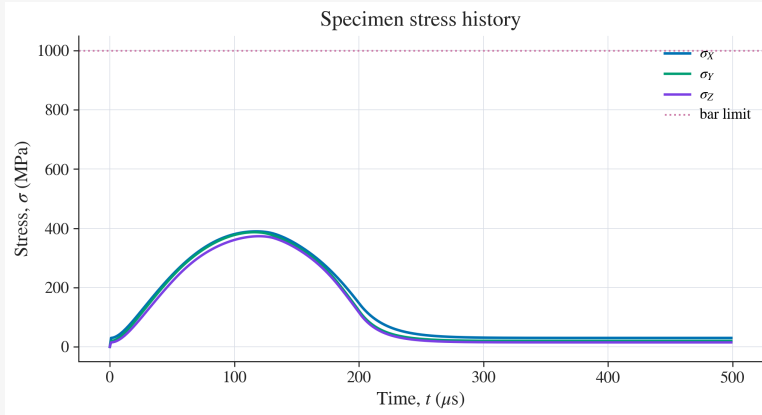
EM pulses on X, X+Y, or X+Y+Z; per-axis amplitudes, duration 200 μ s.

Static pre-stress: $\sigma_x = 30$, $\sigma_y = 20$, $\sigma_z = 15$ MPa. Y/Z delays are programmable from 0–500 μ s; the six-branch damage comparison uses $\Delta t_y = 80$ μ s and $\Delta t_z = 160$ μ s.

Interpretation

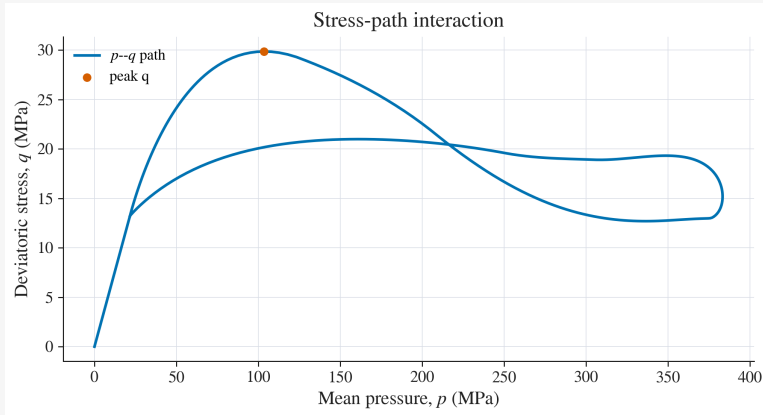
This mode is for stress-path design: changing the pulse delays changes the order in which the material sees mean pressure and deviatoric stress.

Mode 5: stress history



- X, Y, and Z dynamic stresses are all active, so the loading state evolves in three dimensions.
- This setup can be run synchronously or with non-zero Y/Z delays to separate the peaks in time.

Mode 5: stress path



- The curved p - q trajectory is the key diagnostic, not a simple axial stress-strain curve.
- Peak q marks the strongest deviatoric demand before the path relaxes toward high mean pressure.

Mode 6: electromagnetic symmetric multidirectional impact

Use when:

- Hydrostatic or near-hydrostatic dynamic loading is required.
- Specimen stress must exceed one-sided bar limits.
- Momentum cancellation is part of the design requirement.

Key idea

Opposing pulses double specimen stress while keeping single-bar stress unchanged.

$$\sigma_i(t) = \sigma_{i0} + 2A_i g(t), \quad F_i^+ + F_i^- \approx 0.$$

The factor 2 comes from opposed equal pulses; F_i^+ and F_i^- are opposing bar forces.

Mode 6 setup conditions

Material and geometry

Sandstone preset: $E = 15$ GPa, $UCS = 80$ MPa, $\nu = 0.20$, $\rho = 2650$ kg m⁻³.

Specimen: $50 \times 50 \times 50$ mm.

Loading setup

Opposed EM pulses on $\pm X, \pm Y, \pm Z$; peak 400 MPa per bar, duration 200 μ s.

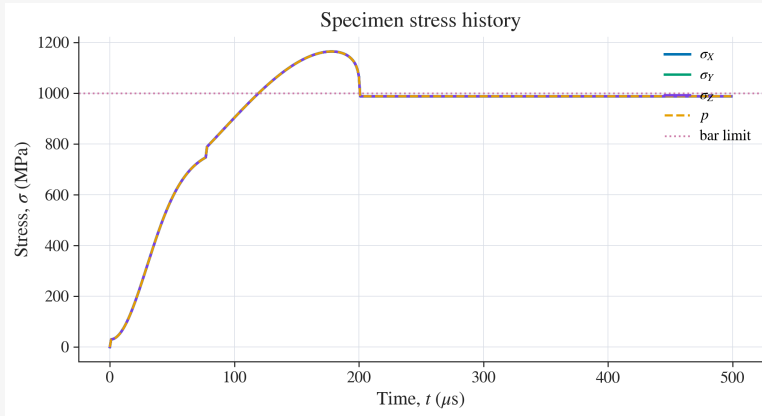
Full XYZ hydrostatic pre-stress:

$p_0 = \sigma_x = \sigma_y = \sigma_z$ (default example $p_0 = 30$ MPa).

Interpretation

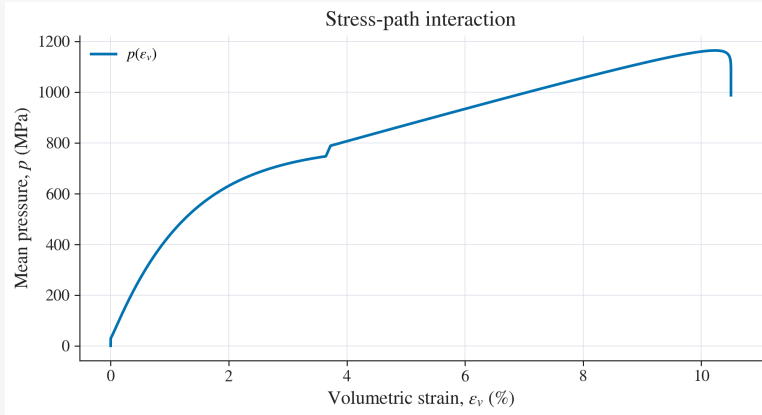
Symmetric opposed loading increases specimen pressure while cancelling net specimen momentum, making it the closest default case to dynamic compaction.

Mode 6: stress history



- Opposed pulses double the specimen-side loading contribution without exceeding the single-bar pulse scale.
- The three principal stresses rise together, so deviatoric stress is reduced relative to Mode 5.

Mode 6: hydrostatic compaction path



- The diagnostic changes from axial strength to pressure–volumetric strain compaction.
- Use this mode when hydrostatic cap response is the target rather than shear failure.

Six-Branch Similarities and Differences

Six-branch comparison: what is shared and what changes

Mode	Drive	Boundary state	What changes in the result	Shared basis
1	Gas X	Unconfined	Baseline axial strength	Sandstone, 50 mm
2	Gas X	Chamber, $\sigma_2 = \sigma_3$	Axisymmetric pressure strengthening	Same gas speed as Mode 1
3	Gas X	Active true triaxial	Independent lateral stresses shift Lode path	Same X impact family
4	EM X	Static triaxial pre-stress	Half-sine amplitude-duration control	EM family, topology branch
5	EM X/Y/Z	Timed one-sided multi-axis pulses	Loading order bends the stress path	Same EM A, τ basis as Mode 4
6	EM $\pm$ axes	Symmetric squeeze; linked p_0 for full XYZ	Hydrostatic compaction and force cancellation	Same EM A, τ basis

Common experimental basis: sandstone preset $E = 15$ GPa, UCS = 80 MPa, $\nu = 0.20$, $\rho = 2650$ kg m⁻³; specimen 50 × 50 × 50 mm.

Six-branch result summary: similarities and differences

Mode	Same as others	Different setup condition	Result signature
1	Same sandstone and specimen geometry	No confinement; gas-gun X at 20 m s^{-1}	Lowest-pressure baseline; only axial branch is meaningful
2	Same gas-gun speed as Modes 1 and 3	Chamber confinement, $\sigma_1 = 30$, $\sigma_2 = \sigma_3 = 20 \text{ MPa}$	Axisymmetric confinement strengthens the specimen
3	Same X impact family as Mode 2	True triaxial pre-stress, 30/20/15 MPa	Lode-angle effect appears through unequal lateral stresses
4	Same EM family as Modes 5 and 6	One EM X pulse, 400 MPa peak, $200 \mu\text{s}$; 30/20/15 MPa	Pulse amplitude and duration are explicit inputs
5	Same one-sided EM reduction basis as Mode 4	Programmable X/Y/Z amplitudes and Y/Z timing	Curved path shows loading-order and path dependence
6	Same EM pulse scale as Modes 4 and 5	Opposed symmetric pulses; full XYZ uses hydrostatic p_0	Hydrostatic compaction dominates; momentum cancels

Focused comparison: traditional, true-triaxial, and symmetric loading

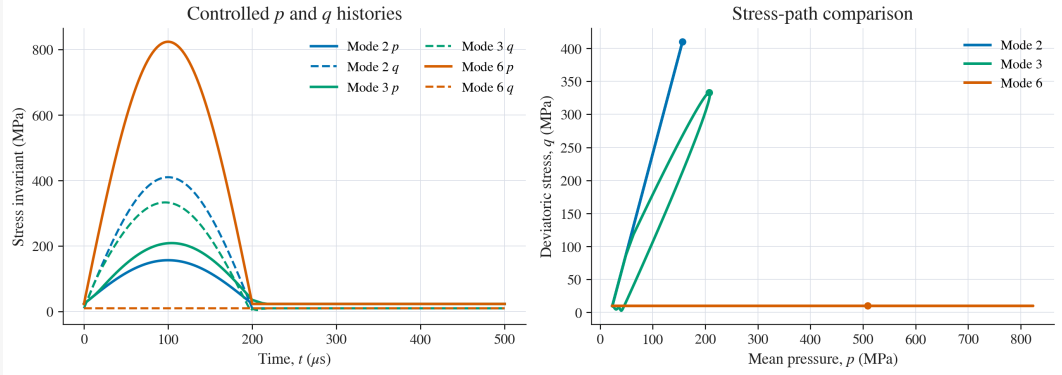
Normalised input for the focused comparison

Same sandstone specimen and half-sine stress-wave scale $A = 400$ MPa, $\tau = 200$ μ s. Mode 6 full XYZ uses a hydrostatic p_0 pre-stress, while Modes 2–3 keep their chamber/Tri-HB baselines.

Mode	Controlled boundary condition	Peak p	Peak q	Peak F	Step 4 result
2	Locked Y/Z chamber	156.7 MPa	410.0 MPa	5.65	$D_f = 1.00$, $D_c = 0.29$
3	True-triaxial lateral response	208.9 MPa	333.1 MPa	3.89	$D_f = 1.00$, $D_c = 0.38$
6	Symmetric XYZ doubled	823.3 MPa	10.0 MPa	0.35	$D_f = 0.00$ in shear law

Here $F = q/q_f$ is the Step 3 failure index used by the Step 4 scalar damage law. Mode 6 stores much more volumetric energy, but it does not activate this deviatoric damage trigger.

Focused comparison: stress-path data



- Mode 2 gives the largest deviatoric demand because the lateral stresses are fixed while X rises.
- Mode 3 reduces q by allowing a true-triaxial lateral response under the same X pulse.
- Mode 6 drives pressure strongly but keeps q small, so it should be interpreted as a cap/compaction case.

Mode selection: match the apparatus to the question

Mode	Best question	Primary diagnostic
1	Baseline dynamic strength	Three-wave equilibrium
2	Axisymmetric confinement	Confinement correction and $\theta = 0$
3	In-situ true triaxial pre-stress	Independent $\sigma_1, \sigma_2, \sigma_3$
4	Clean rate sweep	Half-sine amplitude-duration control
5	Loading-order effects	Δt^* , p - q - θ path
6	Hydrostatic cap response	Symmetric energy folding and momentum cancellation

Practical rule

Choose the simplest mode that still preserves the stress state and timing physics required by the research question.

Experimental Workflow and Stress-Path Reduction

Experimental workflow: same sample, comparable inputs

Controlled experiment rule

Use the same sandstone sample geometry and material constants across the six physics branches. When a branch has pre-static stress, use the default laboratory values from the integrated analysis setup.

Wave-input rule

- Gas-gun modes use the same reference striker condition.
- EM branches use the same half-sine amplitude-duration concept.
- Mode 5 may change Y/Z delay up to 500 μs to expose path dependence.

Step 1 defines the shared experimental state

Inputs

- material preset and specimen geometry
- bar cross-section and elastic properties
- loading family, EM topology, pulse source, and pre-stress
- simulator or uploaded gauge data

Data handoff

Shared keys

setup conditions
reduced stress and wave histories

These keep later steps synchronised with the current setup.

Step 1 equations: bar signals to specimen response

Three-wave stress

$$\sigma_s(t) = \frac{A_b E_b}{2A_s} [\varepsilon_I(t) + \varepsilon_R(t) + \varepsilon_T(t)].$$

One-wave stress

$$\sigma_s(t) = \frac{A_b E_b}{A_s} \varepsilon_T(t).$$

Definitions

- $\varepsilon_I, \varepsilon_R, \varepsilon_T$: incident, reflected, transmitted strains.
- Raw gauge time preserves travel delays; aligned time feeds reduction.
- A_b, A_s, E_b, σ_s : areas, bar modulus, specimen stress.

Strain-rate reduction

$$\dot{\varepsilon}_s(t) = -\frac{2C_0}{L_s} \varepsilon_R(t), \quad \varepsilon_s(t) = \int_0^t \dot{\varepsilon}_s(\tau) d\tau.$$

Bar-wave energy

$$W_i(t) = E_b A_b C_0 \int_0^t \varepsilon_i^2(\tau) d\tau.$$

Definitions

- C_0 : elastic wave speed in the bar.
- L_s : specimen loading length.
- W_i : energy carried by wave branch i .
- τ : dummy integration time, not pulse duration here.

Step 2 equations: wave timing and superposition

Elastic wave speeds

$$M = \frac{E(1-\nu)}{(1+\nu)(1-2\nu)}, \quad c_p = \sqrt{\frac{M}{\rho}}, \quad c_s = \sqrt{\frac{G}{\rho}}.$$

Timing scales

$$t_{\text{travel}} = \frac{L_s}{c_p}, \quad \Delta t^* = \frac{\Delta t}{t_{\text{travel}}}.$$

Definitions

- E, ν, ρ : specimen Young's modulus, Poisson's ratio and density.
- M : constrained P-wave modulus; G : shear modulus.
- c_p, c_s : P- and S-wave speeds in the specimen.

Centre superposition

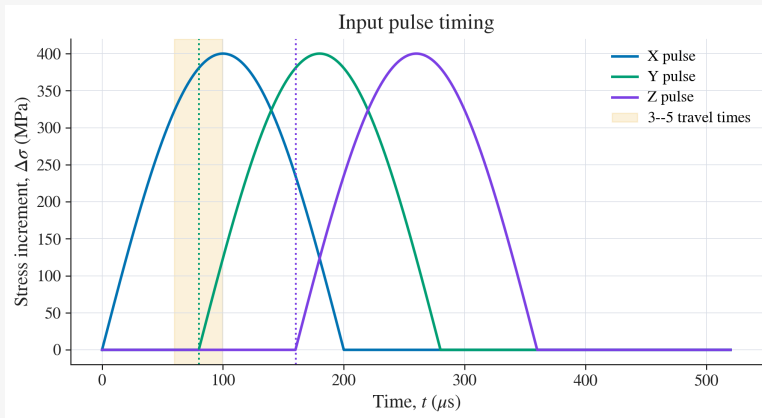
$$\sigma_{\text{centre}}(t) = \sigma_{x0} + \sum_i A_i g\left(t - \Delta t_i - \frac{L_s}{2c_p}\right).$$

$$\eta_{\text{sup}} = \frac{\max_t \sigma_{\text{centre}}(t) - \sigma_{x0}}{A_x}.$$

Definitions

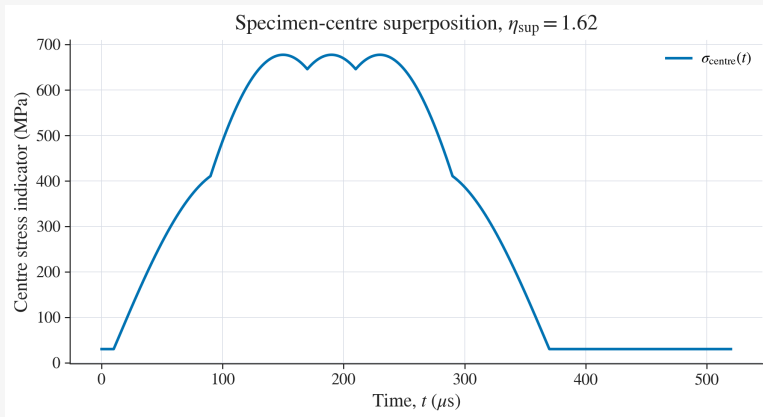
- A_i : pulse amplitude on axis i .
- $g(\cdot)$: selected pulse shape, for example half-sine or Hann.
- Δt_i : firing delay for axis i .
- η_{sup} : superposition factor at specimen centre.

Step 2: input dynamic pulses



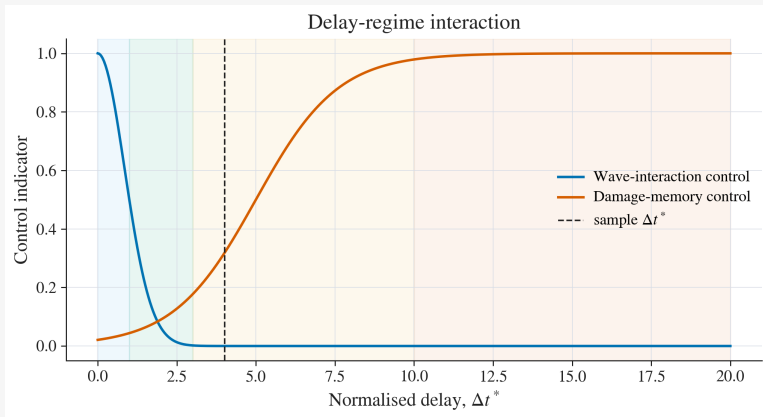
- Pulse amplitude, duration, and delay define the wave-interaction problem before failure is evaluated.
- This plot confirms whether the selected family/topology is one-axis, multi-axis, or symmetric.

Step 2: centre superposition



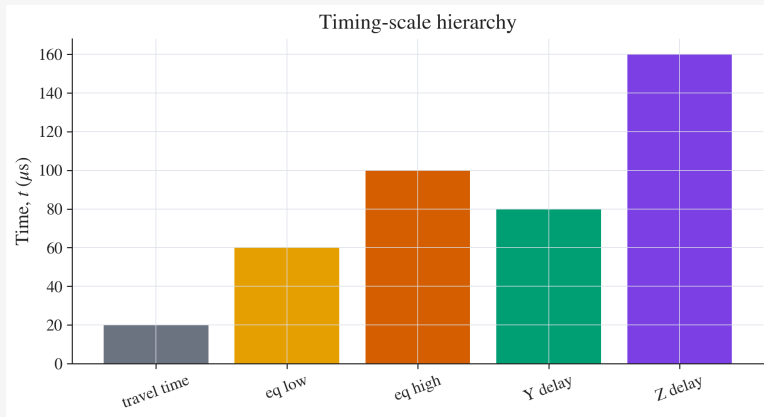
- The centre stress is the sum of delayed wave arrivals at the specimen middle.
- Peak superposition identifies the most severe instant for the later stress-path check.

Step 2: delay regime map



- Small Δt^* gives overlapping pulses; large Δt^* gives sequential loading.
- The regime label explains whether Mode 5 behaves like simultaneous or staged triaxial impact.

Step 2: timing scales



- Travel time normalises the imposed pulse delay so different specimen sizes can be compared.
- The plot links material wave speed, specimen length, and pulse timing in one check.

Step 3 equations: stress invariants and failure index

Stress invariants

$$p = \frac{\sigma_x + \sigma_y + \sigma_z}{3},$$

$$q = \sqrt{\frac{(\sigma_x - \sigma_y)^2 + (\sigma_y - \sigma_z)^2 + (\sigma_z - \sigma_x)^2}{2}}.$$

$$\cos(3\theta) = \frac{3\sqrt{3}}{2} \frac{J_3}{J_2^{3/2}}.$$

Definitions

- $\sigma_x, \sigma_y, \sigma_z$: principal stresses aligned with the bar axes.
- p : mean pressure; q : deviatoric stress magnitude.
- J_2, J_3 : deviatoric stress invariants.
- θ : Lode angle describing compression/extension

Failure envelope

$$h(\theta) = 1 + a_\theta (1 - \cos 3\theta),$$

$$q_f(p, \theta) = (A + Bp^n) h(\theta), \quad F(t) = \frac{q(t)}{q_f(t)}.$$

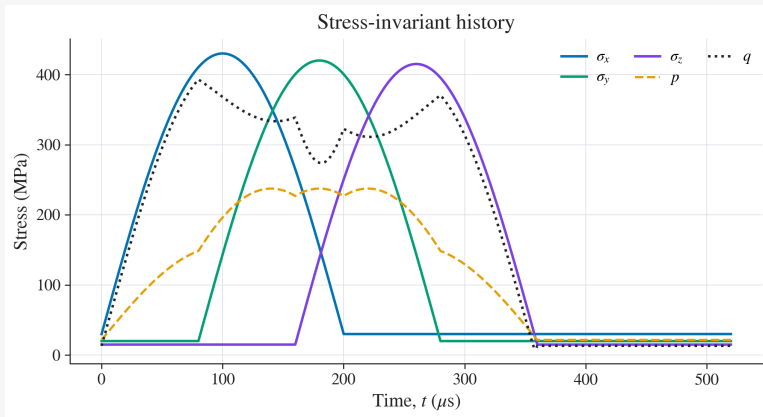
Equivalent strain rate

$$\dot{\epsilon}_{eq} = \sqrt{\frac{2}{3} \cdot \frac{1}{2} [(\dot{\epsilon}_x - \dot{\epsilon}_y)^2 + (\dot{\epsilon}_y - \dot{\epsilon}_z)^2 + (\dot{\epsilon}_z - \dot{\epsilon}_x)^2]}.$$

Definitions

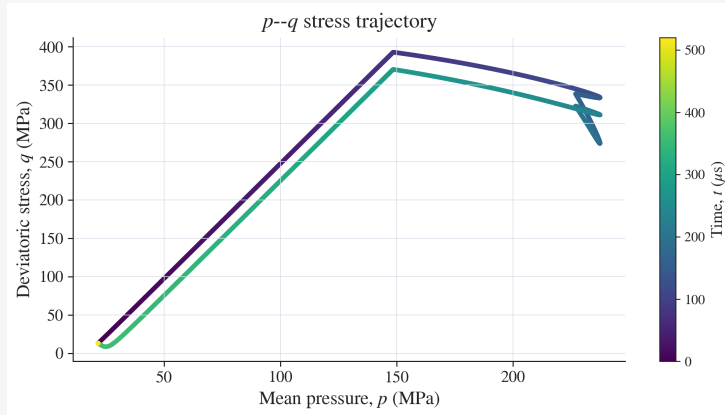
- A, B, n, a_θ : calibrated failure-envelope parameters.
- q_f : available deviatoric strength at the current p, θ .

Step 3: total stresses and invariants



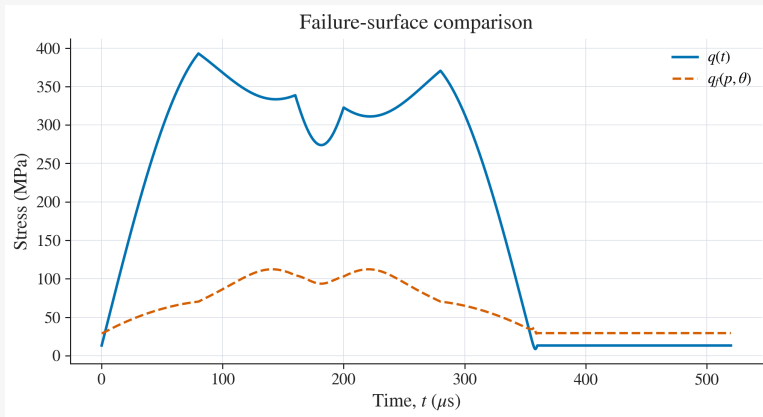
- Total stresses include both static pre-stress and dynamic pulse contributions.
- The derived p , q , and θ histories are the bridge from apparatus loading to failure analysis.

Step 3: p - q trajectory



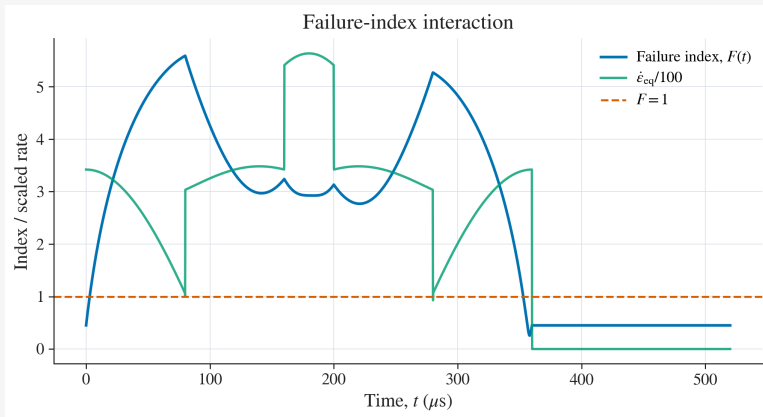
- The trajectory shows whether loading is mostly confinement increase, shear increase, or both.
- Peak q is the critical point for comparing the stress path with the failure envelope.

Step 3: failure surface comparison



- The path is judged against $q_f(p, \theta)$ rather than a single uniaxial strength value.
- Lode-angle correction changes the allowable deviatoric stress for the same mean pressure.

Step 3: failure index and strain rate



- Damage is only allowed to grow when $F = q/q_f$ exceeds one.
- The strain-rate history controls how rapidly overstress is converted into damage in Step 4.

Damage-Model Results and Validation

Step 4 equations: damage evolution

Damage trigger

Damage grows only when the stress state leaves the failure envelope:

$$F(t) = \frac{q(t)}{q_f(t)} > 1, \quad \langle x \rangle = \max(x, 0).$$

Definitions

- $F = q/q_f$: demand/capacity ratio from Step 3.
- q : current deviatoric stress; q_f : failure-envelope value.
- $\langle x \rangle$: Macaulay bracket, so negative overstress gives zero damage growth.
- D : scalar damage variable, from intact (0) to failed (1).

Rate-sensitive law

$$\dot{D} = \frac{(1-D)^\alpha}{\tau_D} R_F R_{\dot{\epsilon}},$$

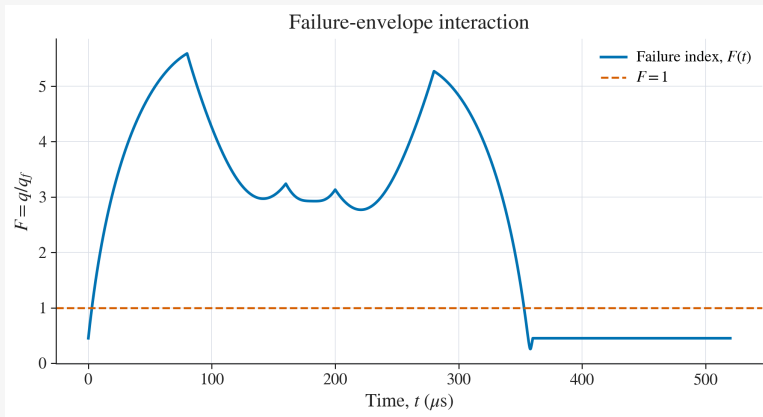
$$R_F = \left\langle \frac{F-1}{F_0} \right\rangle^m, \quad R_{\dot{\epsilon}} = \left(\frac{|\dot{\epsilon}_{eq}|}{\dot{\epsilon}_0} \right)^\beta.$$

$$E(D) = E_0(1-D), \quad c_p(D) = c_{p0} \sqrt{\max(1-D, 0)}.$$

Definitions

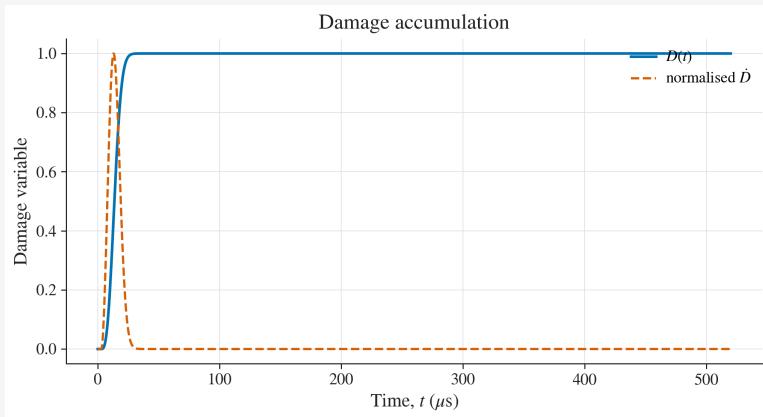
- τ_D : damage time scale; α : saturation exponent.
- F_0, m : overstress normalisation and exponent.
- $\dot{\epsilon}_0, \beta$: reference rate and rate sensitivity.
- $E(D), c_p(D)$: degraded stiffness and P-wave speed.

Step 4: damage trigger



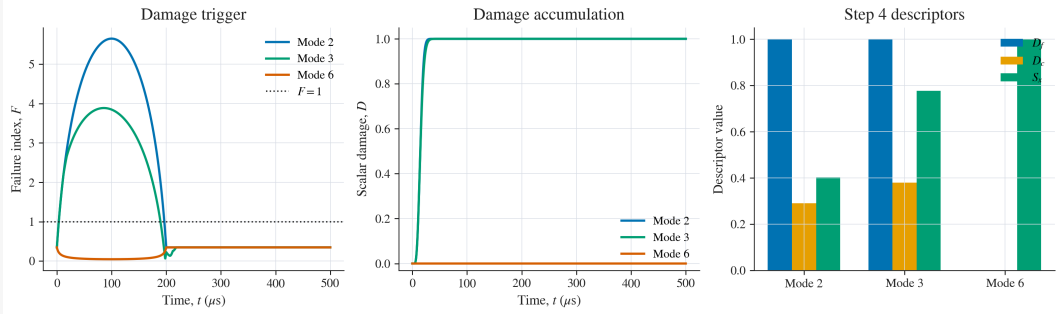
- The horizontal line at $F = 1$ is the onset threshold for damage evolution.
- Time spent above the threshold, not only the peak value, controls the accumulated damage.

Step 4: cumulative damage



- Damage grows during the overstress window and then saturates as stiffness is lost.
- The final value provides the scalar damage state used by the validation descriptors.

Step 4: Modes 2, 3, and 6 damage-model comparison



- With the same pre-stress and pulse scale, Modes 2 and 3 both exceed $F = 1$ and saturate the scalar shear-damage law.
- Mode 3 has a lower peak failure index than Mode 2 because lateral dynamic response reduces the deviatoric contrast.
- Mode 6 remains below the deviatoric trigger; use cap, compaction, or pore-collapse descriptors to validate its damage state.

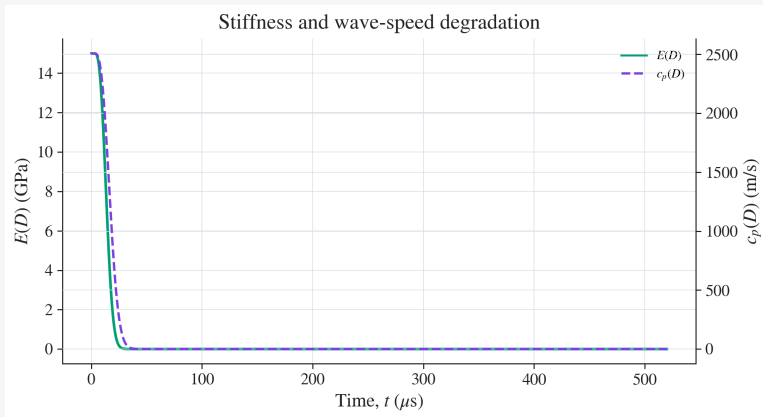
Step 4: controlled damage descriptors

Mode	Damage driver	Onset time	Final D_f	D_c	S_x
2	Axial deviatoric overstress	$3.5 \mu s$	1.00	0.29	0.40
3	Deviatoric overstress plus lateral response	$3.5 \mu s$	1.00	0.38	0.78
6	Hydrostatic/cap response; shear trigger inactive	—	0.00	0.00	1.00

Interpretation

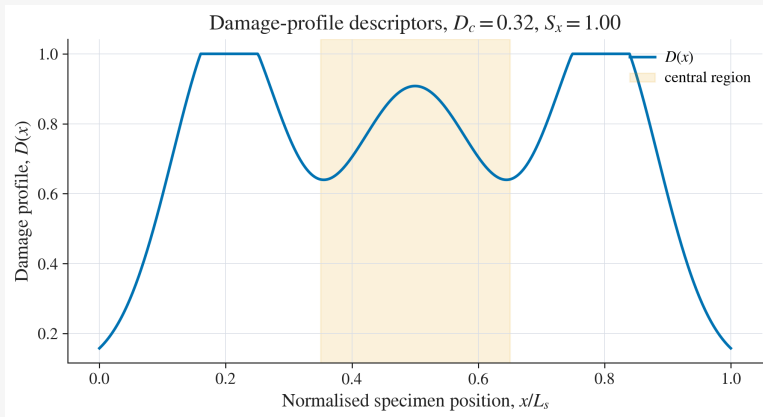
The same Step 4 scalar law is appropriate for Modes 2–3 shear failure comparison. Mode 6 needs an additional volumetric cap-damage measure because high p with low q can crush pores or compact grains without crossing the deviatoric envelope.

Step 4: stiffness and wave-speed loss



- $E(D)$ and $c_p(D)$ translate scalar damage into measurable stiffness and wave-speed degradation.
- These curves are useful for comparing simulated damage with ultrasonic or post-test measurements.

Step 4: damage profile descriptors



- The profile distinguishes central damage from end effects and boundary artefacts.
- Symmetry indicators help identify whether asynchronous loading biased the failure zone.

Step 4 equations: energy and validation descriptors

Energy density

$$W_{\text{el}} = \frac{1}{2E} \left[\sigma_x^2 + \sigma_y^2 + \sigma_z^2 - 2\nu(\sigma_x\sigma_y + \sigma_y\sigma_z + \sigma_z\sigma_x) \right],$$
$$W_{\text{input}}(t) = \int_0^t (\sigma_x \dot{\epsilon}_x + \sigma_y \dot{\epsilon}_y + \sigma_z \dot{\epsilon}_z) d\tau.$$

Definitions

- W_{el} : recoverable elastic energy density.
- W_{input} : cumulative mechanical work density.
- $\sigma_i \dot{\epsilon}_i$: power density contribution from axis i .
- Units are MJ m^{-3} when stresses are MPa and strains are dimensionless.

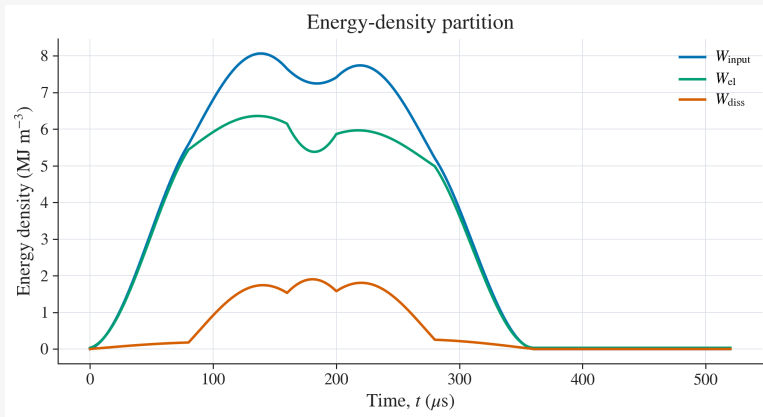
Damage descriptors

$$D_c = \frac{\int_{\text{centre}} D(x) dx}{\int_0^{L_s} D(x) dx},$$
$$S_x = 1 - \frac{|D_{\text{left}} - D_{\text{right}}|}{D_{\text{left}} + D_{\text{right}}}.$$

Definitions

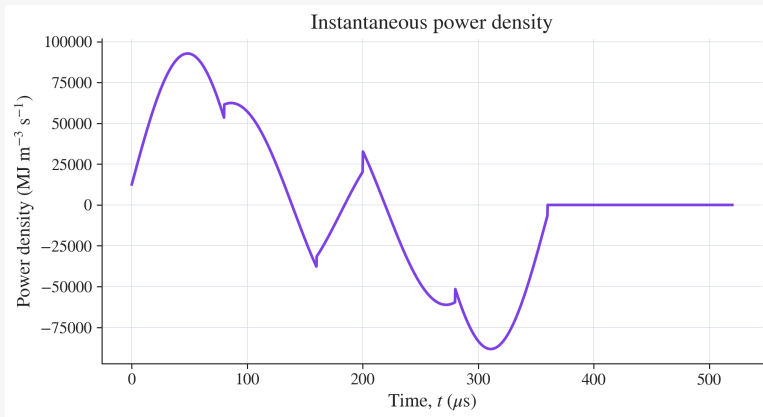
- $D(x)$: estimated damage profile along specimen length.
- D_c : fraction of total damage in the central region.
- $D_{\text{left}}, D_{\text{right}}$: integrated damage on each half.
- $S_x = 1$ means left/right damage symmetry; smaller values mean localisation bias.

Step 4: energy-density histories



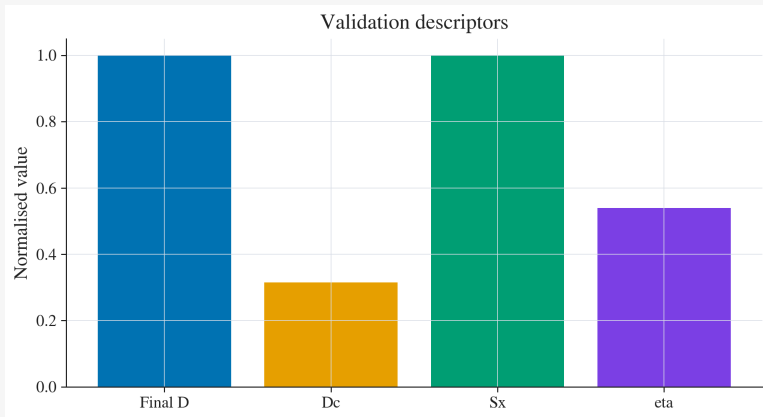
- Input work, recoverable energy, and dissipated energy should evolve consistently with the damage law.
- A large dissipated fraction indicates irreversible cracking or compaction rather than elastic storage.

Step 4: power density



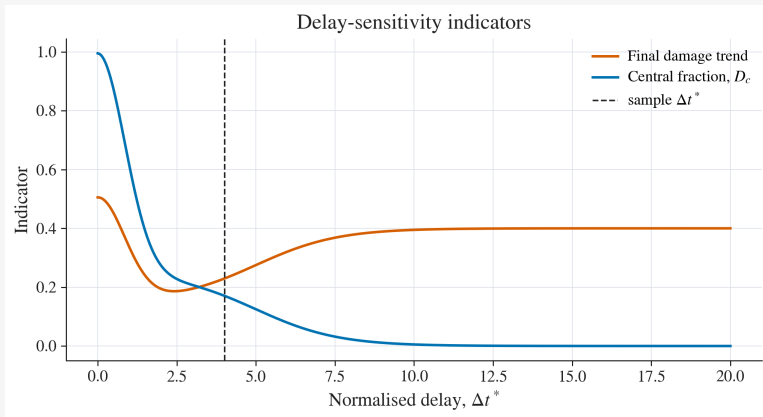
- Power density identifies the short time window where mechanical work is injected most rapidly.
- Compare the peak with the damage-onset time to check whether failure is rate-driven.

Step 4: validation descriptors



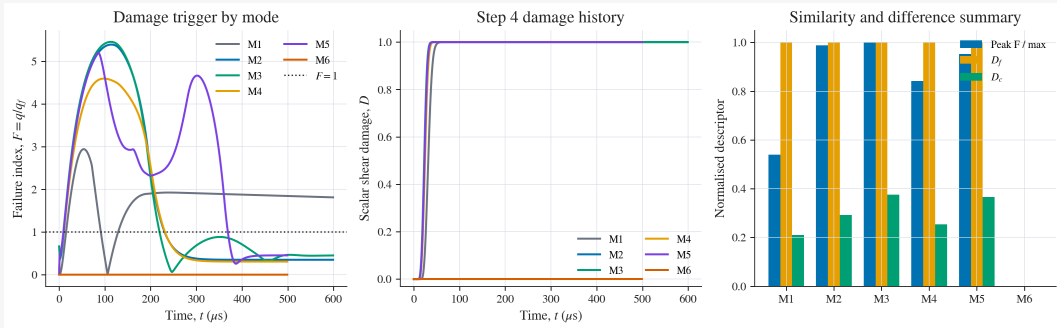
- The descriptors compress a spatial damage field into centre concentration and left/right symmetry.
- They are intended for comparison with DEM fracture maps or recovered specimen observations.

Step 4: delay-sensitivity indicators



- Delay sensitivity links Step 2 timing to the final damage pattern and energy dissipation.
- Use it to choose whether a multi-axis test should be simultaneous or intentionally staggered.

Step 4: six-branch damage-model result



- Modes 1–5 activate the deviatoric shear-damage law under the controlled examples.
- Mode 5 uses the same EM amplitude and duration as Modes 4 and 6, but delayed Y/Z pulses show path dependence.
- Mode 6 is nearly hydrostatic, so the shear-damage trigger stays inactive and a cap/compaction descriptor is needed.

Step 4: six-branch damage descriptors

Mode	Controlled experimental input	Peak F	Onset	D_f	D_c
1	$V = 20 \text{ m s}^{-1}$; no static pre-stress	2.97	$9.0 \mu\text{s}$	1.00	0.21
2	$V = 20 \text{ m s}^{-1}$; 30/20/20 MPa	5.45	$6.0 \mu\text{s}$	1.00	0.29
3	$V = 20 \text{ m s}^{-1}$; 30/20/15 MPa	5.52	$6.0 \mu\text{s}$	1.00	0.38
4	EM X; $A = 400 \text{ MPa}$, $\tau = 200 \mu\text{s}$; 30/20/15 MPa	4.59	$10.0 \mu\text{s}$	1.00	0.25
5	EM XYZ; same A , τ ; Y/Z delays 80/160 μs	5.39	$10.0 \mu\text{s}$	1.00	0.37
6	EM $\pm$ XYZ; same A , τ per opposed bar; hydrostatic p_0	0.00	–	0.00	0.00

Main message

The six physics branches use comparable material inputs, but they do not create the same failure mechanism: Modes 1–5 drive shear failure, while Mode 6 requires volumetric compaction validation.

Outputs connect the model to experiments

Exportable data

- stress–strain histories
- strain-rate histories
- incident/reflected/transmitted energies
- Step 2–4 CSV/XLSX summaries

Validation targets

- bar gauge amplitude and delay
- DIC localisation
- CT damage volume and crack orientation
- DEM bond breakage and dissipated energy

Closing: What The Lab Can Demonstrate

What this reorganised presentation now demonstrates

- Classical SHPB theory is introduced before the modified apparatus modes.
- Mode 2 is presented as the traditional confinement-chamber extension.
- Mode 3 is presented as the true-triaxial, in-situ stress extension.
- Modes 4–6 are grouped as electromagnetic loading for controlled amplitude, duration, timing, and symmetry.
- The comparison and damage results use the same rock sample basis and aligned calibration around $\dot{\epsilon}_{\text{ref}} = 10^{-5} \text{ s}^{-1}$.

Teaching sequence

1. start with Mode 1 wave reduction
2. add confinement through Modes 2–3
3. move to EM pulse control in Modes 4–6
4. finish with Step 2–4 damage validation

Research sequence

1. choose loading family/topology from the stress-path question
2. check bar plasticity and equilibrium
3. export reduced histories
4. calibrate damage and DEM descriptors

Rock dynamic testing becomes a stress-path problem.

The six physics branches turn uniaxial strength, in-situ stress, blast-like pulses, seismic timing, and excavation disturbance into comparable laboratory evidence.